Journal of the American Statistical Association

Taylor & Francis
Taylor & Francis Group

ISSN: 0162-1459 (Print) 1537-274X (Online) Journal homepage: www.tandfonline.com/journals/uasa20

Model-Free Statistical Inference on High-Dimensional Data

Xu Guo, Runze Li, Zhe Zhang & Changliang Zou

To cite this article: Xu Guo, Runze Li, Zhe Zhang & Changliang Zou (2025) Model-Free Statistical Inference on High-Dimensional Data, Journal of the American Statistical Association, 120:549, 186-197, DOI: 10.1080/01621459.2024.2310314

To link to this article: https://doi.org/10.1080/01621459.2024.2310314

JOURNAL OF THE AMERICAN STATISTICAL ASSOCIATION
2025, VOL. 120, NO. 549, 186–197: Theory and Methods
https://doi.org/10.1080/01621459.2024.2310314

Model-Free Statistical Inference on High-Dimensional Data

Xu Guo[a], Runze Li[b], Zhe Zhang[b], and Changliang Zou[c]

[a]School of Statistics, Beijing Normal University, Beijing, China; [b]Department of Statistics, The Pennsylvania State University, University Park, PA; [c]School of Statistics and Data Science, Nankai University, Tianjin, China

ABSTRACT
This article aims to develop an effective model-free inference procedure for high-dimensional data. We first reformulate the hypothesis testing problem via sufficient dimension reduction framework. With the aid of new reformulation, we propose a new test statistic and show that its asymptotic distribution is χ^2 distribution whose degree of freedom does not depend on the unknown population distribution. We further conduct power analysis under local alternative hypotheses. In addition, we study how to control the false discovery rate of the proposed χ^2 tests, which are correlated, to identify important predictors under a model-free framework. To this end, we propose a multiple testing procedure and establish its theoretical guarantees. Monte Carlo simulation studies are conducted to assess the performance of the proposed tests and an empirical analysis of a real-world dataset is used to illustrate the proposed methodology. Supplementary materials for this article are available online including a standardized description of the materials available for reproducing the work.

ARTICLE HISTORY
Received February 2022
Accepted January 2024

KEYWORDS
False discovery rate control; Marginal coordinate hypothesis; Orthogonality; Sufficient dimension reduction.

1. Introduction

High-dimensional data are frequently collected in a large variety of areas such as biomedical imaging, functional magnetic resonance imaging, tomography, tumor classifications, and finance. Many regularization methods have been proposed for simultaneous estimation and variable selection (Fan et al. 2020). Statistical inference for high-dimensional data receives considerable attention in the recent literature. Zhang and Zhang (2014) proposed the debiased Lasso, a low-dimensional projection approach to construct confidence interval and perform hypothesis testing, for linear regression models. Van de Geer et al. (2014) and Javanmard and Montanari (2014) proposed the desparsified Lasso under the setting of generalized linear models, and Ning and Liu (2017) proposed a decorrelated score method that can be applied to a more general family of penalized M-estimators. The debiased Lasso, desparsified Lasso and the decorrelated score method share the same spirit. See, for example, chap. 7 of Fan et al. (2020) for detailed discussion. Fang, Ning, and Liu (2017) generalized the decorrelation method for the low dimensional hypothesis testing approach in high-dimensional proportional hazards models. Fang, Ning, and Li (2020) extended the decorrelated score method to longitudinal data with ultrahigh-dimensional predictors. Shi et al. (2019) developed the constrained partial regularization method for testing linear hypotheses in high-dimensional generalized linear models. Sun and Zhang (2021) proposed a modified profile likelihood-based statistic for hypothesis testing without penalizing the parameters of interest. Aforementioned statistical inference procedures were developed for a specific model such as linear model, generalized linear model or Cox's model. This work aims to develop a model-free statistical inference procedure for high-dimensional data.

In the initial stage of high-dimensional data modeling, it may be quite challenging in correctly specifying a regression model, if not impossible. Furthermore there is lack of effective procedures to validate specified model assumption in the high-dimensional setting. Thus, it is of great interest to develop statistical inference procedures for high-dimensional data without a pre-specified parametric model. To be precise, let $Y \in \mathbb{R}$ be the response variable along with predictor vector $\boldsymbol{X} = (X_1, \ldots, X_p)^\top \in \mathbb{R}^p$. Denote $F(Y|\boldsymbol{X})$ to be the conditional distribution of Y given $\boldsymbol{X}$. The active set $\mathcal{A}$ is defined as

$$\mathcal{A} = \{j : F(Y|\boldsymbol{X}) \text{ functionally depends on } X_j\}.$$

Then $\mathcal{A}^c$, the complement of $\mathcal{A}$, is the index set consisting of all irrelevant predictors. Let $\boldsymbol{X}_{\mathcal{A}} = \{X_j, j \in \mathcal{A}\}$ denote the vector containing all the active predictors. Thus, Y is independent of $\boldsymbol{X}$ given $\boldsymbol{X}_{\mathcal{A}}$, denote it by $Y \perp\!\!\!\perp \boldsymbol{X}|\boldsymbol{X}_{\mathcal{A}}$. In this article, we assume that the set $\mathcal{A}$ is unique. Candes et al. (2018) made a deep discussion about the uniqueness of $\mathcal{A}$. The distribution of $\boldsymbol{X}$ should be nondegenerate otherwise the active set $\mathcal{A}$ may be ambiguous. It is of interest to test whether a predictor X_j contributes to the response Y or not given the other predictors. This can be formulated as testing the hypothesis whether $H_{0j} : j \in \mathcal{A}^c$.

In this article, we propose a new model-free test for H_{0j} in the presence of ultrahigh-dimensional predictors. We first develop a new reformulation of H_{0j} by using statistical framework of sufficient dimension reduction (SDR, Li 2018). Based on this

CONTACT Runze Li ril4@psu.edu Department of Statistics, The Pennsylvania State University, University Park, PA.
The authors contribute equally and are listed in alphabetical order.
Supplementary materials for this article are available online. Please go to www.tandfonline.com/r/JASA.

reformulation, we further propose a test statistic for H_{0j} and show that it converges to χ^2 distribution whose degree of freedom does not depend on unknown population quantities. We also show that the newly proposed test statistic can retain power under local alternative hypotheses.

Under the model-free framework, we study the false discovery rate (FDR) control in the large scale simultaneous testing problem: $H_{0j} : j \in \mathcal{A}^c$ for all $j = 1, \ldots, p$. Classical FDR control procedures including Benjamini-Hochberg (BH) procedure (Benjamini and Hochberg 1995) and Storey procedure (Storey 2002) are typically built on the independence assumption. Other FDR control procedures such as Benjamini-Yekutieli (BY) procedure (Benjamini and Yekutieli 2001) require positive dependence structure and tend to be overly conservative (Cai et al. 2019). These procedures may fail to control the FDR in the presence of complex and strong dependence structure in practice. Liu (2013) developed FDR control procedures for multiple z-tests under Gaussian graphical model setting by assuming the number of pairs of strongly correlated tests bounded. This method can be viewed as a truncated version of BH method, but it cannot be directly extended to the underlying setting of this article because of complicated dependence structure. We propose a new FDR control procedure by using the maximum correlation coefficient to measure the dependence of two tests, and show that the proposed procedure asymptotically controls the FDR at nominal level under mild conditions.

This article makes contribution to both the areas of high dimensional inference and SDR. To the best of our knowledge, it is new to introduce the framework of SDR to make high dimensional inference. Our article illustrates the efficiency of SDR in high dimensional inference.

Our major contributions include the following four aspects.

- First, we investigate high-dimensional inference including testing individual hypothesis and large scale simultaneous testing problem in a model-free framework. Compared with recently popular model-X knockoff procedure (Candes et al. 2018), we achieve FDR control under weaker conditions on the predictors, and we can also determine the significance of individual predictor.
- Second, for the marginal coordinate hypothesis H_{0j} (Cook 2004), this is the first paper which allows the predictors to be ultrahigh-dimensional. Existing papers in the area of SDR focus on the fixed dimension scenario. The asymptotic distributions of existing test statistics are generally in the form of weighted sum of χ^2 distributions with unknown weights. While our proposed test statistic asymptotically follows standard χ^2 distribution.
- Third, our procedures can determine the importance of an individual predictor in ultrahigh-dimensional setting without imposing a specific functional form on regression model. This desirable feature is significantly different from the existing works based on specific parametric models such as linear model or generalized linear model.
- Lastly, we develop new techniques and results to establish the FDR control procedure's theoretical properties. For instance, we establish new results on the tail probability of χ^2 distribution with $h \geq 2$ degrees of freedom in Lemmas S.4–S.5. The maximum correlation coefficient is introduced to quantify the dependence. These techniques and results may be interesting in their own rights and can be useful for other high-dimensional inference problems.

The article is organized as follows. In Section 2, we present a new reformulation. In Section 3, we propose a new test procedure and investigate its asymptotic properties under the null and local alternative hypotheses. In Section 4, we investigate the FDR control procedure which is important to identify active predictors. We present numerical studies in Section 5 and conclusions in Section 6. Proofs and additional simulations are given in the Supplement.

2. A Reformulation via Sufficient Dimension Reduction

We now reformulate the hypothesis $H_{0j} : j \in \mathcal{A}^c$ by using concepts related to sufficient dimension reduction (SDR). Note that H_{0j} is called marginal coordinate hypothesis in the seminal work of Cook (2004), in which the author evaluated the predictor's effect in a model-free setting by adopting the SDR framework (Li 2018). Following Cook (2004), several authors have developed test procedures for H_{0j} (Shao, Cook, and Weisberg 2007; Yu and Dong 2016; Dong, Yang, and Yu 2016) when the dimension p is fixed. These tests have a major limitation that the asymptotic distributions of these test statistics are sums of weighted χ^2 distributions with unknown weights. This makes their practical implementation challenging.

Define the central subspace $\mathcal{S}_{Y|\boldsymbol{X}}$ to be the minimum subspace $\mathcal{S}$ given by the column space of a $p \times d$ matrix $\boldsymbol{B} = (\boldsymbol{\beta}_1, \ldots, \boldsymbol{\beta}_d)$ with $d < p$ such that $Y \perp\!\!\!\perp \boldsymbol{X} | \boldsymbol{B}^\top \boldsymbol{X}$. Under some mild conditions, the central subspace uniquely exists and contains all information of $F(Y|\boldsymbol{X})$ (Li 2018). Throughout this article, it is assumed that $\mathcal{S}_{Y|\boldsymbol{X}}$ exists and its dimension d is fixed.

Based on the concepts of the central subspace, we can reformulate H_{0j}. To this end, we first consider the estimation of the central subspace $\mathcal{S}_{Y|\boldsymbol{X}}$ because the basis matrix $\boldsymbol{B}$ is unknown. Without loss of generality, assume that $\mathbb{E}(\boldsymbol{X}) = \boldsymbol{0}$, and $\boldsymbol{\Sigma} = \mathrm{var}(\boldsymbol{X}) > 0$.

We impose the following least squares condition (LSC): for any function $f(Y)$,

$$\boldsymbol{\Sigma}^{-1}\mathrm{cov}(\boldsymbol{X}, f(Y)) \in \mathcal{S}_{Y|\boldsymbol{X}}.$$

We now discuss several sufficient conditions of the LSC.

The first sufficient condition is the commonly imposed linearity condition (LC):

$$\mathbb{E}[\boldsymbol{X}|\boldsymbol{B}^\top\boldsymbol{X}] \text{ is a linear function of } \boldsymbol{B}^\top\boldsymbol{X}.$$

The LC is satisfied when $\boldsymbol{X}$ follows an elliptical distribution. This LC is widely adopted in the literature of SDR. For recent adoption, see Qian, Ding, and Cook (2019), Ding, Qian, and Wang (2020), Luo (2022), Zeng, Mai, and Zhang (2024), and Ying and Yu (2022).

Instead of assuming the LC, recently Guan, Xie, and Zhu (2017) established that even when $\boldsymbol{X}$ follows the mixture multivariate skew-elliptical distributions, the LSC can still hold. See Corollary 2 in Guan, Xie, and Zhu (2017). Furthermore, the LSC also holds for a model-based SDR method called principal fitted components (PFC) introduced by Cook and Forzani

(2008). Actually Cook and Forzani (2008) assumed an inverse regression model:

$$\boldsymbol{X} = \mathbf{AB}[\mathbf{f}(Y) - \mathbb{E}\mathbf{f}(Y)] + \mathbf{e},$$

where $\mathbf{f}(Y) \in \mathbb{R}^h$ is some user-specified functions of Y, the matrix $\mathbf{B} \in \mathbb{R}^{d\times h}$ is of rank $d < h$ and the matrix $\mathbf{A} \in \mathbb{R}^{p\times d}$ is semi-orthogonal. The error term $\mathbf{e}$ is independent of Y and normally distributed. Cook and Forzani (2008) showed that under the inverse regression model, the LSC holds; see Zeng, Mai, and Zhang (2024) for more details.

When the LSC is not valid, one may consider to make a transformation on data before applying the proposed test procedure in practice. In S.4.6 in the supplement, we conduct some numerical studies on this issue. Our numerical results imply that the proposed test may fail when the LSC is not satisfied. We further demonstrate marginal Gaussian transformation (Mai, He, and Zou 2022) may be helpful to improve the performance of the proposed test when the LSC violates.

The LSC enables us to obtain several directions in $\mathcal{S}_{Y|X}$ by choosing a series of transformations on Y: $f_1(Y), \dots, f_h(Y)$, with pre-specified $h > d$. There are various choices for the transformation functions $f_k(Y)$. The original SIR (Li 1991) corresponds to choosing the slice indicator function $f_k(Y) = I(Y \in J_k)$. Here $\{J_1, \dots, J_h\}$ is a partition of the support of Y. Fung et al. (2002) suggested choosing the normalized B-spline basis functions for $f_k(Y)$; Yin and Cook (2002) suggested the normalized polynomial transformation $f_k(Y) = Y^k$ up to power h; Cook and Ni (2006) recommended choosing $f_k(Y) = YI(Y \in J_k)$. For the PFC method, Cook and Forzani (2008) and Zeng, Mai, and Zhang (2024) considered $f_k(Y) = Y^k$ with $h = 3$. See also Yin and Li (2011) for a summary. It is challenging in deriving optimal transformation functions because the exact form of alternative hypothesis is unknown.

Define

$$\boldsymbol{\beta}_k^0 = \arg\min_{\boldsymbol{\beta}_k} \mathbb{E}[\{f_k(Y) - \boldsymbol{X}^\top \boldsymbol{\beta}_k\}^2], \quad \text{for} \quad k = 1, \dots, h.$$

Denote $\boldsymbol{B}_0 = (\boldsymbol{\beta}_1^0, \dots, \boldsymbol{\beta}_h^0)$. Then $\text{Span}(\boldsymbol{B}_0) \subseteq \mathcal{S}_{Y|X}$. Following the literature of SDR, we take one step further by assuming the coverage condition (CC)

$$\text{Span}(\boldsymbol{B}_0) = \mathcal{S}_{Y|X},$$

whenever $\text{Span}(\boldsymbol{B}_0) \subseteq \mathcal{S}_{Y|X}$ so that the subspace spanned by $\boldsymbol{B}_0$ coincides with the central subspace. This condition is reasonable. Actually the larger h is, the $\text{Span}(\boldsymbol{B}_0)$ is larger. Thus, when h is large, $\text{Span}(\boldsymbol{B}_0)$ could be very close to $\mathcal{S}_{Y|X}$. This condition is also widely adopted in the literature of SDR. See for instance Yu and Dong (2016), Qian, Ding, and Cook (2019), Ding, Qian, and Wang (2020), Weng and Yin (2022), and Yu, Yao, and Xue (2022). As discussed by Yin and Cook (2002), if the conditional distribution of Y given $\boldsymbol{X}$ depends only on conditional moments up to the dth, then the CC holds. In the location-scale regression model, the d is simply equal to 2. Under the inverse regression model assumption, the CC holds for the PFC method. For a full discussion about the coverage condition, see Li (2018).

For $j = 1, \dots, p$, let $\boldsymbol{b}_j^\top = (\beta_{1j}^0, \dots, \beta_{hj}^0)$ be the jth row of $\boldsymbol{B}_0$, where β_{kj}^0 is the jth element of $\boldsymbol{\beta}_k^0 \in \mathbb{R}^p, k = 1, \dots, h$. Note that if $j \in \mathcal{A}^c$, then any of the h linear combinations $\boldsymbol{X}^\top \boldsymbol{\beta}_k^0, k = 1, \dots, h$ must not involve X_j. Thus, $Y \perp\!\!\!\perp \boldsymbol{X} | \boldsymbol{X}_{\mathcal{A}}$ and $Y \perp\!\!\!\perp \boldsymbol{X} | \boldsymbol{B}_0^\top \boldsymbol{X}$ imply that $\sum_{k=1}^h |\beta_{kj}^0| > 0$ for $j \in \mathcal{A}$, and $\sum_{k=1}^h |\beta_{kj}^0| = 0$ for $j \in \mathcal{A}^c$. Thus, the following proposition are valid.

Proposition 1. Suppose that the LSC and CC conditions hold, we have: $H_{0j} : j \in \mathcal{A}^c$ if and only if $H'_{0j} : \boldsymbol{b}_j = \boldsymbol{0}$.

If the CC does not hold, then we have $H_{0j} : j \in \mathcal{A}^c$ implies $H'_{0j} : \boldsymbol{b}_j = \boldsymbol{0}$. That is, H'_{0j} is a necessary null hypothesis of H_{0j}. When H'_{0j} is rejected, we can also reject H_{0j}. This reformulation converts the original hypothesis testing problem into a parametric hypothesis without knowing the specific form of the conditional distribution of Y given $\boldsymbol{X}$. We next construct a score-type statistic for testing H'_{0j}.

3. A New Test Statistic and its Limiting Distribution

In this section, we develop a test statistic for H'_{0j} to achieve the goal of testing H_{0j}. To get the insight of proposed test for H'_{0j}, we first consider the low dimensional setting in which p is fixed. By the definition of $\boldsymbol{\beta}_k^0$, we may consider using $\mathbb{E}[X_j\{f_k(Y) - \boldsymbol{X}^\top \boldsymbol{\beta}_k^0\}]$ (i.e., the derivative of $\mathbb{E}[\{f_k(Y) - \boldsymbol{X}^\top \boldsymbol{\beta}_k^0\}^2]$ with respect to β_{kj}^0), for $k = 1, \dots, h$ to construct a test statistic. This is similar to the score test in likelihood setting. Under H'_{0j},

$$\mathbb{E}[X_j\{f_k(Y) - \boldsymbol{X}^\top \boldsymbol{\beta}_k^0\}] = \mathbb{E}[X_j\{f_k(Y) - \boldsymbol{Z}_j^\top \boldsymbol{\gamma}_{kj}\}],$$

where $\boldsymbol{Z}_j$ is the subvector of $\boldsymbol{X}$ without X_j, and $\boldsymbol{\gamma}_{kj}$ is the subvector of $\boldsymbol{\beta}_k^0$ without β_{kj}^0.

Suppose that $\{\boldsymbol{X}_i, Y_i\}$, $i = 1, \dots, n$, are random samples generated from $\{\boldsymbol{X}, Y\}$. Similarly, we denote $\boldsymbol{Z}_{ij}$ for the sample of $\boldsymbol{Z}_j$. By the definitions of $\boldsymbol{\beta}_k^0$, we then obtain the least squares estimate of $\boldsymbol{\gamma}_{kj}$, $\widehat{\boldsymbol{\gamma}}_{kj}$, by a linear regression of $f_k(Y)$ on $\boldsymbol{Z}_j$. Then it is natural to consider the score-type test statistic:

$$T_{nk}^j = \frac{1}{\sqrt{n}} \sum_{i=1}^n X_{ij}(f_k(Y_i) - \boldsymbol{Z}_{ij}^\top \widehat{\boldsymbol{\gamma}}_{kj}).$$

It can be shown that under the null hypothesis, $(T_{n1}^j, \dots, T_{nh}^j)$ jointly converges to a multivariate normal distribution with zero mean.

We now consider high-dimensional setting. The test procedure does not work in high-dimensional setting since the least squares estimate is not well defined. However, we may consider the score type test T_{nk}^j by replacing the least squares estimate with the corresponding penalized estimate $\widehat{\boldsymbol{\gamma}}_{kj}$ of $\boldsymbol{\gamma}_{kj}$. In high-dimensional setting, it is not uncommon to impose sparsity assumption on regression coefficients. That is, only a small subset of the predictors are significant to the response variable. We estimate $\boldsymbol{\gamma}_{kj}$ by its penalized least squares estimate

$$\widehat{\boldsymbol{\gamma}}_{kj} = \arg\min \frac{1}{2n} \sum_{i=1}^n \{f_k(Y_i) - \boldsymbol{Z}_{ij}^\top \boldsymbol{\gamma}_{kj}\}^2 + \sum_{l=1}^{p-1} p_{\lambda_Y}(|\gamma_{kj,l}|), \quad k = 1, \dots, h, \tag{3.1}$$

where $p_{\lambda_Y}(\cdot)$ is a penalty function with a tuning parameter λ_Y.

Remark 1. For multiple testing problem to be studied in next section, we need to calculating all $\widehat{\boldsymbol{\gamma}}_{kj}$, $j = 1, \ldots, p$ and $k = 1, \ldots, h$. This requires to solve hp regularized optimization problems, and leads to expensive computational cost. For multiple testing problem, we recommend to estimate $\boldsymbol{\beta}_k^0$ by using

$$\widehat{\boldsymbol{\beta}}_k = \arg\min \frac{1}{2n}\sum_{i=1}^{n}\{f_k(Y_i) - \boldsymbol{X}_i^\top \boldsymbol{\beta}_k\}^2 + \sum_{l=1}^{p} p_{\lambda_Y}(|\beta_{kl}|),$$
$$k = 1, \ldots, h. \tag{3.2}$$

Then set $\widehat{\boldsymbol{\gamma}}_{kj}$ as the subvector of $\widehat{\boldsymbol{\beta}}_k$ without $\widehat{\beta}_{kj}$. This can save computational cost dramatically. This still has the theoretical guarantee.

The test statistic T_{nk}^j can be further refined since it could fail in some situation. To see this, note that

$$\begin{aligned} T_{nk}^j &= \frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_{ij}(f_k(Y_i) - \boldsymbol{Z}_{ij}^\top \widehat{\boldsymbol{\gamma}}_{kj}) \\ &= \frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_{ij}(f_k(Y_i) - \boldsymbol{Z}_{ij}^\top \boldsymbol{\gamma}_{kj}) + \frac{(\boldsymbol{\gamma}_{kj} - \widehat{\boldsymbol{\gamma}}_{kj})^\top}{\sqrt{n}}\sum_{i=1}^{n} \boldsymbol{Z}_{ij}X_{ij}. \end{aligned}$$

Although under mild conditions, the first term converges to a normal distribution, the second term may not be asymptotically normal. Indeed, for the penalized estimate of $\boldsymbol{\gamma}_{kj}$, it may not be always valid without strong condition on $\boldsymbol{\gamma}_{kj}$, such as minimal signal condition (Fan et al. 2020), to ensure that $\sqrt{n}(\widehat{\boldsymbol{\gamma}}_{kj} - \boldsymbol{\gamma}_{kj})$ converges to normal distribution. Moreover, the Euclidean norm of the term $n^{-1}\sum_{i=1}^{n} \boldsymbol{Z}_{ij}X_{ij}$ can diverge since its dimension is p. Thus, it is of importance to refine the test statistic T_{nk}^j so that it performs well in general settings. We next employ the idea of orthogonalization to refine T_{nk}^j. The orthogonalization plays a critical role in reducing the bias from the estimators of high-dimensional nuisance parameters (Belloni, Chernozhukov, and Kato 2015; Ning and Liu 2017; Belloni et al. 2018).

Instead of employing $\mathbb{E}[X_j\{f_k(Y) - \boldsymbol{Z}_j^\top \boldsymbol{\gamma}_{kj}\}]$ in high-dimensional setting, we propose using

$$\mathbb{E}\Big[(X_j - \boldsymbol{Z}_j^\top \boldsymbol{\theta}_j^*)\{f_k(Y) - \boldsymbol{Z}_j^\top \boldsymbol{\gamma}_{kj}\}\Big] \tag{3.3}$$

to construct a test statistic for H_{0j}', where $\boldsymbol{\theta}_j^* = \mathbb{E}[\boldsymbol{Z}_j\boldsymbol{Z}_j^\top]^{-1}\mathbb{E}[\boldsymbol{Z}_jX_j]$. This is because the term in (3.3) possesses the orthogonality property:

$$\begin{aligned} \frac{\partial}{\partial \boldsymbol{\gamma}_{kj}}\mathbb{E}\Big[(X_j - \boldsymbol{Z}_j^\top \boldsymbol{\theta}_j^*)\{f_k(Y) - \boldsymbol{Z}_j^\top \boldsymbol{\gamma}_{kj}\}\Big] &= \mathbb{E}\left[\boldsymbol{Z}_j\left(X_j - \boldsymbol{Z}_j^\top \boldsymbol{\theta}_j^*\right)\right] \\ &= 0 \end{aligned}$$

under some conditions. Even if $\boldsymbol{\gamma}_{kj}$ is estimated at a slower rate than $n^{-1/2}$, such orthogonalization can still guarantee zero mean with introduction and suitable estimation of $\boldsymbol{\theta}_j^*$. The orthogonalization makes the test statistic for the parameter of interest locally insensitive to the high dimensional nuisance parameters $\boldsymbol{\gamma}_{kj}$'s.

In order to reduce the influence of nuisance parameters, we estimate $\boldsymbol{\theta}_j^*$ by a penalized least squares estimate

$$\widehat{\boldsymbol{\theta}}_j = \arg\min \frac{1}{2n}\sum_{i=1}^{n}(X_{ij} - \boldsymbol{Z}_{ij}^\top \boldsymbol{\theta}_j)^2 + \sum_{l=1}^{p-1} p_{\lambda_X}(|\theta_{j,l}|), \tag{3.4}$$

where $p_{\lambda_X}(\cdot)$ is a penalty function with a tuning parameter λ_X.

Define $\boldsymbol{S}_i^j = (S_{i1}^j, \ldots, S_{ih}^j)^\top$ with $S_{ik}^j = (X_{ij} - \boldsymbol{Z}_{ij}^\top \widehat{\boldsymbol{\theta}}_j)(f_k(Y_i) - \boldsymbol{Z}_{ij}^\top \widehat{\boldsymbol{\gamma}}_{kj})$ for each k. By the orthogonality property of $\mathbb{E}\Big[(X_j - \boldsymbol{Z}_j^\top \boldsymbol{\theta}_j^*)\{f_k(Y) - \boldsymbol{Z}_j^\top \boldsymbol{\gamma}_{kj}\}\Big]$, it is natural to construct a test statistic based on

$$\bar{\boldsymbol{S}}_n^j = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \boldsymbol{S}_i^j.$$

That is, $\bar{\boldsymbol{S}}_n^j = (\bar{S}_{n1}^j, \ldots, \bar{S}_{nh}^j)^\top$ with $\bar{S}_{nk}^j = n^{-1/2}\sum_{i=1}^{n}(X_{ij} - \boldsymbol{Z}_{ij}^\top \widehat{\boldsymbol{\theta}}_j)(f_k(Y_i) - \boldsymbol{Z}_{ij}^\top \widehat{\boldsymbol{\gamma}}_{kj})$.

As shown in the Appendix, $\bar{\boldsymbol{S}}_n^j$ asymptotically follows a normal distribution with mean 0 and covariance matrix $\boldsymbol{\Omega}_j$, where the (l,k)-element of $\boldsymbol{\Omega}_j$ is $\mathbb{E}[\eta_l^j\eta_k^j(X_j - \boldsymbol{Z}_j^\top \boldsymbol{\theta}_j^*)^2]$ for $l, k = 1, \ldots, h$, where $\eta_k^j = f_k(Y) - \boldsymbol{Z}_j^\top \boldsymbol{\gamma}_{kj}$, $k = 1, \ldots, h$. Thus, we propose the following test statistic for H_{0j}'

$$W_{nj} = \bar{\boldsymbol{S}}_n^{j\top}\widehat{\boldsymbol{\Omega}}_j^{-1}\bar{\boldsymbol{S}}_n^j,$$

where $\widehat{\boldsymbol{\Omega}}_j$ is a consistent estimate of $\boldsymbol{\Omega}_j$ with (k,l)-elements $\widehat{\omega}_{kl}$ being

$$\frac{1}{n}\sum_{i=1}^{n}(f_l(Y_i) - \boldsymbol{Z}_{ij}^\top \widehat{\boldsymbol{\gamma}}_{lj})(f_k(Y_i) - \boldsymbol{Z}_{ij}^\top \widehat{\boldsymbol{\gamma}}_{kj})(X_{ij} - \boldsymbol{Z}_{ij}^\top \widehat{\boldsymbol{\theta}}_j)^2.$$

Let s_Y be the sparsity level for $\boldsymbol{\gamma}_{kj}$, $k = 1, \ldots, h$. That is, $s_Y = \max_{1\le j\le p, 1\le k\le h} \|\boldsymbol{\gamma}_{kj}\|_0$. Also denote the sparsity level for the parameter $\boldsymbol{\theta}_j^*$ to be $s_X = \max_{1\le j\le p} \|\boldsymbol{\theta}_j^*\|_0$. Before we introduce technical conditions, let us introduce some new notations related to sub-Gaussian and sub-exponential random variables.

Definition 1. (i) A random variable X is sub-Gaussian if its tail probability satisfies $\Pr\{|X| \geqslant t\} \leqslant 2\exp\left(-t^2/K_1^2\right)$ for all $t \geqslant 0$. $\|X\|_{\varphi_2} = \inf\{t > 0 : \mathbb{E}\exp(X^2/t^2) \leqslant 2\}$ is the sub-Gaussian norm of X, and is the smallest K_1.

(ii) A random variable X is sub-exponential if its tail probability satisfies $\Pr\{|X| \geqslant t\} \leqslant 2\exp\left(-t/K_2\right)$ for all $t \geqslant 0$. $\|X\|_{\varphi_1} = \inf\{t > 0 : \mathbb{E}\exp(|X|/t) \leqslant 2\}$ is the sub-exponential norm of X, and is the smallest K_2.

The following technical conditions are imposed to establish the asymptotical null distribution of W_{nj}, although they may not be the weakest conditions.

- (B1) $\|\widehat{\boldsymbol{\gamma}}_{kj} - \boldsymbol{\gamma}_{kj}\|_1 = O_p(\lambda_Y s_Y)$, $\|\widehat{\boldsymbol{\theta}}_j - \boldsymbol{\theta}_j^*\|_1 = O_p(\lambda_X s_X)$.
- (B2) η_k^j, $k = 1, \ldots, h$, and X_j, $j = 1, \ldots, p$ are all sub-Gaussian random variables with finite expectation. $\left\|\eta_k^j\right\|_{\psi_2}$ and $\left\|X_j\right\|_{\psi_2}$ are uniformly bounded.
- (B3) $\max\{s_Y, s_X\} = o(\sqrt{n}/\log p)$, $\max\{\lambda_X, \lambda_Y\} = C\sqrt{\log p/n}$.
- (B4) $\boldsymbol{\Sigma} = \mathrm{cov}(\boldsymbol{X}) > 0$, and $\boldsymbol{\Omega}_j > 0$.

These conditions are commonly used in the literature. Condition (B1) is for error bound of the penalized least squares estimators in linear regression models. The penalized least squares estimates with the Lasso, SCAD, and MCP penalties satisfy this condition (Fan et al. 2020). Condition (B2) is imposed for establishing the L_∞ norm of $n^{-1/2}\sum_{i=1}^{n}\eta_{ik}^{j}\boldsymbol{Z}_{ij}$. Sub-Gaussian assumption is a commonly used condition in the literature (Fan et al. 2020). Furthermore, the dimension of p is allowed to be an exponential order of the sample size n according to Conditions (B2) and (B3). The rate for the sparsity levels in Condition (B3) is commonly imposed, see for instance Ning and Liu (2017). Condition (B4) is also a mild condition.

The limiting distribution of W_{nj} under the null hypothesis and local alternative hypotheses are given in the following theorem, whose proof is given in the Appendix.

Theorem 1. Suppose that the LSC and CC conditions, and Conditions (B1)–(B4) are valid.

(i) Under the null hypothesis H_{0j}, it follows that $W_{nj}\to\chi_h^2$ in distribution, where χ_h^2 is the Chi-square distribution with h degrees of freedom;

(ii) Under the alternative hypothesis $H_{1n}:\sqrt{n}\boldsymbol{b}_j\to\boldsymbol{b}\neq\boldsymbol{0}$, it follows that

$$W_{nj}\to\chi_h^2\left(\delta_j^2\boldsymbol{b}^\top\boldsymbol{\Omega}_j^{-1}\boldsymbol{b}\right)$$

in distribution, where $\chi_h^2(a)$ is the noncentral Chi-square distribution with h degrees of freedom and noncentrality parameter a, and $\delta_j=\mathbb{E}(X_j^2)-\mathbb{E}(X_j\boldsymbol{Z}_j^\top)\mathbb{E}(\boldsymbol{Z}_j\boldsymbol{Z}_j^\top)^{-1}\mathbb{E}(\boldsymbol{Z}_jX_j)$.

Theorem 1(a) implies that we are able to test the hypothesis whether $F(Y|\boldsymbol{X})$ functionally depends on the jth predictor or not in ultrahigh-dimensional setting without imposing a specific functional form on regression model. This desirable feature is significantly different from the existing works based on specific parametric model such as linear model or generalized linear model. Further distinguished from most of existing results in the SDR literature on hypothesis testing, our proposed test statistic asymptotically follows standard χ^2 distribution, instead of sums of weighted χ^2 distributions with unknown weights. This makes our procedure easy to implement in practice.

Theorem 1(b) implies that the proposed test W_{nj} can detect the alternative hypotheses which converge to the null at the rate of $n^{-1/2}$. From the proof of Theorem 1(b), we can show that $W_{nj}\to\infty$ for fixed alternative hypothesis $H_1:\boldsymbol{b}_j\neq\boldsymbol{0}$. Thus, W_n has asymptotic power 1 for any fixed alternative hypothesis.

4. False Discovery Rate Control

In Section 3, we propose a test of significance for an individual predictor. In practice, it is of interest to test simultaneously all p hypotheses. That is, $H_{0j}:j\in\mathcal{A}^c$ for all $j=1,\dots,p$. This is potentially useful to identify important features in the high-dimensional data. To avoid spurious discoveries, the FDR control is commonly adopted. Recently, Barber and Candès (2015) introduced a novel knockoff framework to control the FDR under the linear model. Candes et al. (2018) further developed a model-X knockoff procedure that can control FDR without assuming a specific regression model. However, the model-X knockoff procedure requires knowing the joint distribution of the predictors. This makes the knockoff method challenging in practical implementation. See also Barber, Candes, and Samworth (2020). This section aims to develop a FDR control procedure under a model-free framework. The new procedure does not require the knowledge of the joint distribution of the predictors.

For a series of individual hypothesis H_{0j}, we can construct the corresponding test statistics W_{nj}. As shown in the Theorem 1, under null hypothesis W_{nj} follows an asymptotically Chi-squared distribution with degrees of freedom h.

A FDR controlling procedure is to find a threshold t to control the multiple testing effect. For any threshold $t>0$, let $R_0(t)=\sum_{j\in\mathcal{A}^c}I\left(W_{nj}\geqslant t\right)$, and $R(t)=\sum_{j=1}^{p}I\left(W_{nj}\geqslant t\right)$ be the total number of false discoveries and the total number of discoveries associated with t, respectively.

Define false discovery proportion (FDP) and FDR as follows:

$$\mathrm{FDP}(t)=\frac{R_0(t)}{R(t)\vee 1},\quad \mathrm{FDR}(t)=\mathbb{E}\{\mathrm{FDP}(t)\}.$$

Under a pre-specified FDR level α, the ideal choice of the threshold t would be

$$t_0=\inf\{t\in\mathbb{R}:\mathrm{FDP}(t)\leqslant\alpha\}.$$

In practice, we need to estimate $R_0(t)$ since the inactive set is unknown. Intuitively, $R_0(t)$ can be reasonably estimated by $|\mathcal{A}^c|G(t)$, where $G(t)=\Pr(\chi_h^2\geqslant t)$. As a result, $\mathrm{FDP}(t)$ can be estimated by

$$\widehat{\mathrm{FDP}}(t)=\frac{|\mathcal{A}^c|G(t)}{R(t)\vee 1}.$$

Thus, we propose to estimate the threshold level t by

$$\widehat{t}=\inf\left\{0\leqslant t\leqslant b_p:\widehat{\mathrm{FDP}}(t)\leqslant\alpha\right\}.$$

and reject the null hypothesis H_{0j} if $W_{nj}\geqslant\widehat{t}$, where $b_p=2\log p+2d_0\log\log p$ with $2d_0<h-2$. If there is no $t\in[0,b_p]$ such that $\widehat{\mathrm{FDP}}(t)\leqslant\alpha$, we will directly set $\widehat{t}=2\log p+(h-1.5)\log\log p$.

In above, we estimate $R_0(t)$ by $|\mathcal{A}^c|G(t)$ when $t\in[0,b_p]$. In fact under the restriction $2d_0<h-2$ and other mild conditions, we will show

$$\sup_{0\leqslant t\leqslant b_p}\left|\frac{\sum_{j\in\mathcal{A}^c}I\left(W_{nj}\geqslant t\right)}{|\mathcal{A}^c|G(t)}-1\right|\to 0, \tag{4.1}$$

in probability. In other words, it implies that the total number of false discoveries $R_0(t)$ can be consistently estimated by $|\mathcal{A}^c|G(t)$ in this pre-specified region. In practice, $|\mathcal{A}^c|$ is not a prior knowledge, we could approximate $|\mathcal{A}^c|$ by p since $|\mathcal{A}|=o(p)$.

Remark 2. The choice of d_0 depends on the value of h. Since $h-2$ is the boundary value to guarantee the theoretical results, we suggest choosing a slightly smaller value $2d_0=h-2-\epsilon_h$ in practice, where ϵ_h is a small positive constant. It is noteworthy that d_0 is allowed to be negative if $h\leqslant 2$. In Xia, Cai, and Li (2018), they studied the case of asymptotic normal test statistic which corresponds to $h=1$. They set the search region to be $[0,\sqrt{2\log p-2\log\log p}]$, where the right boundary point is smaller than $\sqrt{2\log p}$.

It is crucial to characterize the dependence structure across the tests in order to establish (4.1). Denote

$$\widetilde{\boldsymbol{S}}_n^j = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} \boldsymbol{\eta}_i^j\left(X_{ij} - \boldsymbol{Z}_{ij}^\top\boldsymbol{\theta}_j^*\right), \quad \widetilde{W}_{nj} = \widetilde{\boldsymbol{S}}_n^{j\top}\boldsymbol{\Omega}_j^{-1}\widetilde{\boldsymbol{S}}_n^j.$$

Further denote $\widetilde{\boldsymbol{K}}_n^j = \boldsymbol{\Omega}_j^{-\frac{1}{2}}\widetilde{\boldsymbol{S}}_n^j$. Define the maximum correlation coefficient

$$\rho_{ij}^* = \max_{\|\boldsymbol{a}\|=1,\|\boldsymbol{b}\|=1}\left|\mathrm{corr}\left(\boldsymbol{a}^\top\widetilde{\boldsymbol{K}}_n^i, \boldsymbol{b}^\top\widetilde{\boldsymbol{K}}_n^j\right)\right|$$

which quantifies the dependence between $\widetilde{W}_{ni}$ and $\widetilde{W}_{nj}$. For $i, j \in \mathcal{A}^c$, define

$$\mathcal{B}_1 := \left\{(i,j) : i \neq j, \rho_{ij}^* \geqslant (\log p)^{-2-\epsilon_0}\right\} \text{ for some } \epsilon_0 > 0$$

which contains the pairs (i, j) with high correlation.

In order to ensure the estimated FDR could be controlled at pre-specified level, the difference between constructed tests W_{nj} and $\widetilde{W}_{nj}$ must be uniformly bounded. The following proposition implies that this is valid under mild assumptions

Proposition 2. Suppose that the LSC and CC conditions, and conditions B1, B2, and B4 are valid. Assume $\lambda_X, \lambda_Y = O(\sqrt{\frac{\log p}{n}})$, $\max(s_X, s_Y) = s$. It follows that

(i) $\|\bar{\boldsymbol{S}}_n^j - \widetilde{\boldsymbol{S}}_n^j\|_1 = O_p(F(n,p,s)) = O_p(\frac{\log p}{\sqrt{n}}s)$,
(ii)$\|\widehat{\boldsymbol{\Omega}}_j - \boldsymbol{\Omega}_j\|_2 = O_p(G(n,p,s))$,

hold uniformly for $j \in \mathcal{A}^c$, where

$$G(n,p,s) = \sqrt{\frac{(\log np)^5}{n} \vee \frac{s\log p\log(np)}{n}} \vee s^2\log(np)(\frac{\log p}{n})^{\frac{3}{2}} \vee s^3\log(np)(\frac{\log p}{n})^2.$$

Furthermore, it follows that

$$\max_{j\in\mathcal{A}^c}|W_{nj} - \widetilde{W}_{nj}| = O_p\left(\max\left(\frac{(s\log p)^2}{n}, \frac{s(\log p)^2}{\sqrt{n}}, G(n,p,s)\log p\right)\right).$$

We next show the FDR control result of the proposed procedure in Theorem 2. The following technical conditions are imposed to facilitate its technical proof.

(D1) There exists a positive constant ϵ_1 such that $|\mathcal{B}_1| = O([\log p]^{h-2d_0-2-\epsilon_1})$.
(D2) $\max\left(F(n,p,s), G(n,p,s)\log p\right) = o(1)$.
(D3) $\lambda_{\min}(\boldsymbol{\Omega}_j^{-1}) \geqslant \lambda_0 > 0$ and $\lambda_{\max}(\boldsymbol{\Omega}_j^{-1}) \leqslant \lambda_1$ uniformly in $j = 1, \ldots, p$.
(D4) $\delta_j \geqslant c > 0$ uniformly for j where $\delta_j = \mathbb{E}(X_j^2) - \mathbb{E}(X_j\boldsymbol{Z}_j^\top)\mathbb{E}(\boldsymbol{Z}_j\boldsymbol{Z}_j^\top)^{-1}\mathbb{E}(\boldsymbol{Z}_jX_j)$.

These conditions are mild and commonly used in the literature of FDR. See, for example, Liu (2013), Xia, Cai, and Li (2018) and references therein. Condition D1 requires a bounded number of strongly correlated pairs which correspond to tests in inactive set. The purpose is to control the variance of $R_0(t)$. Condition D2 assumes the difference between constructed test and its ideal version is uniformly bounded with rate $o(1)$, and therefore to ensure the consistency of $\sum_{j\in\mathcal{A}^c} I\left(W_{nj} \geqslant t\right)$ to $|\mathcal{A}^c|G(t)$. Condition D3 assumes the smallest and the largest eigenvalues of $\boldsymbol{\Omega}_j$ are uniformly bounded.

Theorem 2. Under Conditions in Proposition 2 and Conditions D1–D4 with $p \leqslant cn^r$ for some $c > 0$ and $r > 0$. In addition, assume $|\mathcal{A}| = o(p)$. It follows that

$$\limsup_{(n,p)\to\infty} \mathrm{FDR}(\widehat{t}) \leqslant \alpha$$

with probability 1.

In Theorem 1, the dimension p is allowed to grow exponentially fast in n. However, in order to control FDR, it requires $p = O(n^r)$ to apply the moderate deviation result in Liu (2013).

5. Numerical Studies

5.1. Study 1

In this study, we conduct numerical studies to assess the finite sample performance of the proposed procedures. To implement the proposed test, we need to determine the transformation functions $f_k(\cdot)$. In our simulation, we take the $f_k(\cdot), k = 1, \ldots, h$ to be a set of quadratic B-spline bases with $h - 2$ inner knots, which yields a set of h quadratic B-spline bases. An investigation of the impact of transformation functions is given in the supplement. In all simulation studies, we set $p = 2000$, and all simulation results are based on 1000 replications. We consider the Lasso (Tibshirani 1996) and the SCAD (Fan and Li 2001) in the penalized least squares estimate introduced in Section 3. We first study the impact of h on the performance of the proposed test.

5.1.1. The Choice of h

One needs to choose h to implement the proposed procedure. We investigate this issue by Monte Carlo study. To this end, we generate the data from the following three models.

Model I: $Y = X_1 + X_2 + \epsilon$.
Model II: $Y = (X_1 + X_2)/\{0.5 + (1.5 + X_3 + X_4)^2\} + 0.1\epsilon$.
Model III: $Y = 3\sin(X_1) + 3\sin(X_p) + \exp(-2X_3)\epsilon$.

These three models correspond to linear model, nonlinear model and heteroscedastic model, respectively. The dimension of central subspace, d, is 1, 2, and 3, respectively. The predictor vector $\boldsymbol{X} = (X_1, \ldots, X_p)$ is generated from $N(\boldsymbol{0}, \Sigma)$ with the (i,j)-element of Σ being $\sigma_{ij} = 0.5^{|i-j|}$ for $1 \leq i, j \leq p$. The error ϵ is generated from $N(0, 1)$ and independent of $\boldsymbol{X}$. The active sets $\mathcal{A}$ for these models are $\{1, 2\}$, $\{1, 2, 3, 4\}$, and $\{1, 3, p\}$, respectively. We consider sample size $n = 200$ and 400.

To examine the impact of h on the performance of W_{nj}, we set $h = 1, 2, \ldots, 20$ and plot the power curve as a function of h for several js. This enables us to see the overall trend of the power function as h increases. Moreover, we study the impact of choice of penalty function on the performance of test statistics.

Figure 1 presents the empirical power functions with respect to h for several js when the penalty in the penalized least squares is taken to be the Lasso penalty. The results for the SCAD penalty

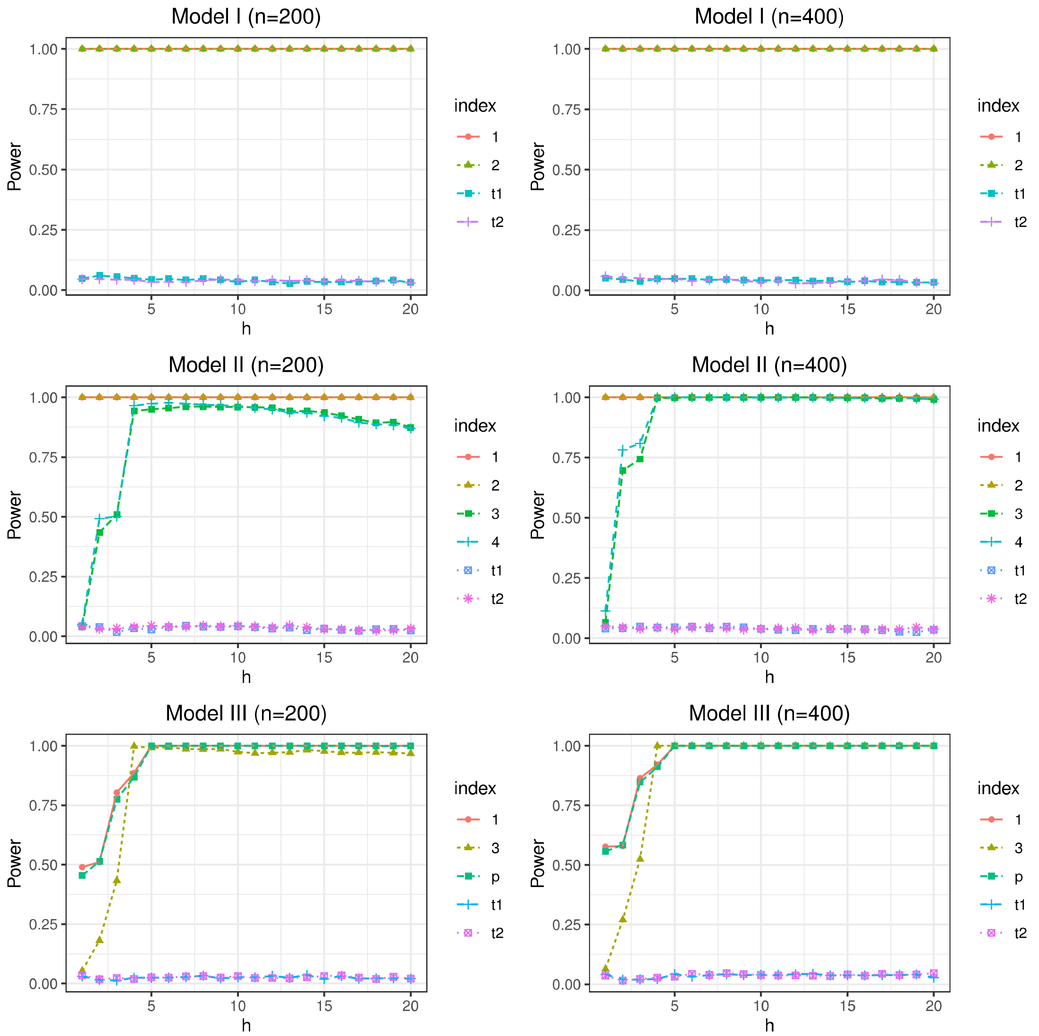

Figure 1. The plot of empirical size and power with respect to h when the Lasso penalty is used. "t1" and "t2" represent randomly selected inactive variables.

are similar to those for the Lasso penalty, and are depicted in Figure S.1 in the supplement. As shown in Figure 1, it can be seen that for truly active variables, the empirical power has a sharp increase as h increases if $h \leqslant d+1$, and remains stable in a wide range of h. It is also observed that the empirical power decreases slowly as h increases when h is much larger than d and $n = 200$. It can also be seen from Figure 1 that power of some active variables in Model II does not approach to 1. In particular, the empirical power of X_3 in Model II achieves its maximum around $h = 5$, but its value is slightly less than 1. This phenomenon disappears when the sample size increases from 200 to 400. Figure 1 also implies that the empirical powers stand firm around 1 when $h > d+1$ for $n = 400$. For inactive variables, the empirical size always stays around nominal level $\alpha = 0.05$. Figure 1 and S.1 seems to imply that $h = 5$ is good choice for Models I, II and III. Thus, we set $h = 5$ in Sections 5.1.2 and 5.1.3.

5.1.2. Comparison with Existing Methods

This section is devoted to comparing the performance of the proposed procedure with two popular methods: the LDPE method (Zhang and Zhang 2014) and the decorrelated score method (Ning and Liu 2017) by pretending all the observations are characterized by a linear model. Let T^{ZZ} and T^{NL} stand for these two test procedures, respectively. We compare the performance of the three procedures in terms of empirical size and power. The proposed test uses quadratic B-spline transformation with $h = 5$.

The data were generated from Models I, II, and III described in Section 5.1.1. We summarize the simulation results in terms of rejection rate, which is defined to be the proportion of p-values being smaller than the nominal level $\alpha = 0.05$ based on 1000 replications. In each simulation, we test H_{0j} for $j = 1, \ldots, p$. For inactive predictors (i.e., X_j with $j \in \mathcal{A}^c$), it is expected that the proportion is close to the nominal level, while for active predictors, it is expected to reject H_{0j} with a large probability, and the rejection rate is the estimated power of the underlying test.

The results are summarized in Tables 1 and S.1 when the Lasso penalty and the SCAD penalty are used, respectively. These two tables only display the rejection rates for predictors X_1-X_5 and X_{1996}-X_{2000} to save space. From these two tables, it can be seen that for linear model (Model I), the newly proposed test performs as well as the other two tests. All three tests control the size well and have good empirical power. For nonlinear model (Model II) and heteroscedastic error model (Model III),

Table 1. Empirical rejection rate of H_{0j} with $\alpha = 5\%$ when the Lasso penalty is used.

n	Method	X_1	X_2	X_3	X_4	X_5	X_{1996}	X_{1997}	X_{1998}	X_{1999}	X_{2000}
Model I: X_1 and X_2 are active predictors											
200	W_n-Lasso	1.000	1.000	0.064	0.043	0.046	0.037	0.037	0.034	0.045	0.037
	T^{NL}-Lasso	1.000	1.000	0.069	0.049	0.050	0.045	0.040	0.046	0.055	0.054
	T^{ZZ}	1.000	1.000	0.141	0.070	0.054	0.043	0.049	0.042	0.053	0.056
400	W_n-Lasso	1.000	1.000	0.065	0.036	0.035	0.042	0.048	0.036	0.039	0.035
	T^{NL}-Lasso	1.000	1.000	0.070	0.050	0.056	0.054	0.048	0.045	0.043	0.052
	T^{ZZ}	1.000	1.000	0.093	0.063	0.060	0.056	0.055	0.048	0.043	0.052
Model II: X_1, X_2, X_3, and X_4 are active predictors											
200	W_n-Lasso	1.000	1.000	0.927	0.956	0.068	0.045	0.041	0.036	0.039	0.034
	T^{NL}-Lasso	1.000	0.999	0.047	0.083	0.067	0.045	0.049	0.043	0.051	0.051
	T^{ZZ}	1.000	1.000	0.041	0.038	0.045	0.049	0.055	0.042	0.057	0.039
400	W_n-Lasso	1.000	1.000	0.997	1.000	0.056	0.037	0.046	0.055	0.034	0.035
	T^{NL}-Lasso	1.000	1.000	0.087	0.161	0.052	0.066	0.046	0.048	0.057	0.054
	T^{ZZ}	1.000	1.000	0.029	0.082	0.040	0.047	0.048	0.047	0.045	0.046
Model III: X_1, X_3, and X_p are active predictors											
200	W_n-Lasso	1.000	0.055	0.988	0.055	0.037	0.029	0.018	0.035	0.056	1.000
	T^{NL}-Lasso	0.304	0.075	0.470	0.070	0.065	0.051	0.052	0.053	0.059	0.272
	T^{ZZ}	0.285	0.078	0.499	0.083	0.052	0.046	0.034	0.055	0.052	0.294
400	W_n-Lasso	1.000	0.046	0.999	0.050	0.045	0.025	0.030	0.031	0.044	1.000
	T^{NL}-Lasso	0.358	0.070	0.478	0.054	0.054	0.047	0.049	0.057	0.039	0.349
	T^{ZZ}	0.358	0.063	0.496	0.064	0.042	0.054	0.040	0.047	0.050	0.343

all three tests also control Type I error at the desired nominal level. But the newly proposed test has much higher power than the other two tests. In particular, the power of the proposed test statistic W_{nj} for Model II is very close to 1 for $H_{03} : 3 \in \mathcal{A}^c$, while the empirical powers of T^{ZZ} and T^{NL} are around 0.05 even when $n = 400$.

5.1.3. Comparison of FDR Control Procedures

We next investigate the finite sample performance of the proposed FDR control procedures by simultaneously testing $H_{0j} : j \in \mathcal{A}^c$ for all $j = 1, \dots, p$. This simulation study is designed to compare the newly proposed FDR control procedure with model-X knockoff procedures (Candes et al. 2018), for which we consider the Lasso coefficient-difference (LCD) statistic.

To understand the impact of sparsity level on FDR control procedures, we consider the following two models, which are modified from Models I and II.

Model IV: $Y = \mathbf{X}^\top \boldsymbol{\beta}_1 + \epsilon$.

Model V: $Y = (\mathbf{X}^\top \boldsymbol{\beta}_2)/\{0.5 + \left(1.5 + X_{p-1} + X_{p-2}\right)^2\} + 0.1\epsilon$.

In our simulation, we set $\boldsymbol{\beta}_1 = (\overbrace{1, \dots, 1}^{s_1}, 0, \dots, 0)$ and $\boldsymbol{\beta}_2 = (\overbrace{1, \dots, 1}^{s_2}, 0, \dots, 0)$. The sparsity levels of Model IV and V are s_1 and $s_2 + 2$, respectively. The sparsity levels of each model are set to be 4, 6, and 8. The distribution of $\mathbf{X}$ and ϵ are the same as those for Models I and II. In this simulation study, $p = 2000$ and the sample size $n = 200$ and 400. For the newly proposed FDR control procedure, the Lasso is also used, and we set $h = 5$. We search the $\widehat{t}$ over $[0, 2\log p + 2.75 \log\log p]$ and set $\widehat{t} = 2\log p + 3.5 \log\log p$ if $\widehat{t}$ does not exist over the interval.

The empirical FDR and empirical power over 1000 replications are presented in Table 2 at the significance level $\alpha = 0.1$ or 0.2, where Power $= |\widehat{\mathcal{A}} \cap \mathcal{A}|/|\mathcal{A}|$ with $\widehat{\mathcal{A}} = \left\{j : W_{nj} \geqslant \widehat{t}\right\}$. From Table 2, it can be seen that for Models IV and V, the proposed procedure and the model-X knockoff procedures with lasso coefficient-difference (LCD) statistic can control the FDR well. For nonlinear model (model V), the proposed method generally has larger powers than the model-X knockoff procedures. The Knockoff+, a more conservative knockoff procedure, has much conservative FDR and therefore has very low power under the setting of Model V.

5.2. Study 2

Based on simulation results in Section 5.1, we can draw the following conclusions. First, a large enough value of h is sufficient to achieve good performance. Second, the choice of penalty function does not appear to have a significant impact on the performance. In this study, we further investigate the impact of different choices of the distributions of $\mathbf{X}$ and the sparsity level s.

5.2.1. Impact of Predictors' Distribution

We first examine how the distribution of $\mathbf{X}$ affects the empirical performance of the proposed procedures. We consider models with d linear indices $U_k = \mathbf{X}^\top \boldsymbol{\beta}_k$, where $k = 1, \dots, d$.

Model I': $Y = U_1 + \epsilon$.
Model II': $Y = \frac{U_1}{0.5 + (1.5 + U_2)^2} + 0.1\epsilon$.
Model III': $Y = \sinh(U_1) + \sinh(U_2) + \sinh(U_3) + 0.5\epsilon$.
Model IV': $Y = U_1 + \sinh(U_2) + 0.3\exp(U_3) + |U_4 + 3| + \epsilon$.
Model V': $Y = U_1 + \sinh(U_2) + \exp(U_3) + |U_4 + 3| + 4\sin(U_5) + \epsilon$.

The coefficient vector corresponding to U_k is denoted by $\boldsymbol{\beta}_k$. We define $\mathcal{M}_k$ as the index set of nonzero components of $\boldsymbol{\beta}_k$, and further assume that $\mathcal{M}_{k_1} \cap \mathcal{M}_{k_2} = \emptyset$ for $k_1 \neq k_2$. In Model II', we have $|\mathcal{M}_1| = \lfloor 0.8s \rfloor$ and $|\mathcal{M}_2| = s - |\mathcal{M}_1|$ for any sparsity

Table 2. Empirical size and power of FDR control procedures at nominal level $\alpha = 0.1,\ 0.2$.

		Model IV				Model V			
		$\alpha = 0.1$		$\alpha = 0.2$		$\alpha = 0.1$		$\alpha = 0.2$	
Sparsity	Method	FDR	Power	FDR	Power	FDR	Power	FDR	Power
					$n = 200$				
	Proposed	0.091	0.999	0.097	0.999	0.080	0.796	0.083	0.797
4	LCD-Knockoff	0.061	1.000	0.154	1.000	0.062	0.500	0.129	0.500
	LCD-Knockoff+	0.008	0.012	0.153	0.480	0.002	0.002	0.065	0.049
	Proposed	0.074	0.997	0.092	0.997	0.064	0.864	0.071	0.865
6	LCD-Knockoff	0.065	1.000	0.155	1.000	0.063	0.638	0.158	0.646
	LCD-Knockoff+	0.023	0.051	0.137	1.000	0.007	0.007	0.152	0.314
	Proposed	0.062	0.974	0.119	0.982	0.066	0.784	0.089	0.798
8	LCD-Knockoff	0.080	1.000	0.164	1.000	0.068	0.628	0.160	0.664
	LCD-Knockoff+	0.063	0.236	0.158	1.000	0.017	0.026	0.151	0.548
					$n = 400$				
	Proposed	0.084	1.000	0.086	1.000	0.064	0.969	0.065	0.969
4	LCD-Knockoff	0.061	1.000	0.151	1.000	0.060	0.500	0.133	0.500
	LCD-Knockoff+	0.016	0.026	0.153	0.482	0	0	0.059	0.046
	Proposed	0.063	1.000	0.078	1.000	0.051	0.982	0.058	0.982
6	LCD-Knockoff	0.067	1.000	0.151	1.000	0.058	0.667	0.151	0.667
	LCD-Knockoff+	0.026	0.060	0.141	1.000	0.004	0.004	0.147	0.321
	Proposed	0.051	1.000	0.103	1.000	0.042	0.985	0.088	0.988
8	LCD-Knockoff	0.078	1.000	0.158	1.000	0.061	0.739	0.151	0.744
	LCD-Knockoff+	0.065	0.257	0.154	1.000	0.016	0.028	0.132	0.735

Table 3. Empirical rejection rate with respect to different distributions of $\boldsymbol{X}$.

	W_n	T^{NL}	W_n					T^{NL}				
Dist	Inactive1	Inactive1	Index1	Index2	Index3	Index4	Index5	Index1	Index2	Index3	Index4	Index5
							Model I′					
Chi	0.046	0.046	1.000	***	***	***	***	1.000	***	***	***	***
Mixture	0.052	0.065	1.000	***	***	***	***	1.000	***	***	***	***
MSN	0.049	0.050	1.000	***	***	***	***	1.000	***	***	***	***
MVN	0.000	0.000	1.000	***	***	***	***	1.000	***	***	***	***
Tdist	0.062	0.036	1.000	***	***	***	***	1.000	***	***	***	***
							Model II′					
Chi	0.041	0.058	1.000	0.993	***	***	***	0.996	0.004	***	***	***
Mixture	0.043	0.040	1.000	1.000	***	***	***	1.000	0.100	***	***	***
MSN	0.048	0.049	1.000	1.000	***	***	***	1.000	0.254	***	***	***
MVN	0.060	0.045	1.000	1.000	***	***	***	1.000	0.056	***	***	***
Tdist	0.040	0.075	1.000	1.000	***	***	***	1.000	0.064	***	***	***
							Model III′					
Chi	0.032	0.051	0.999	1.000	1.000	***	***	0.202	0.209	0.301	***	***
Mixture	0.044	0.047	1.000	1.000	1.000	***	***	0.898	0.676	0.922	***	***
MSN	0.053	0.057	1.000	1.000	1.000	***	***	0.784	0.808	0.973	***	***
MVN	0.052	0.050	1.000	1.000	1.000	***	***	0.836	0.805	0.971	***	***
Tdist	0.029	0.269	1.000	1.000	1.000	***	***	0.385	0.374	0.442	***	***
							Model IV′					
Chi	0.031	0.058	0.770	1.000	1.000	0.219	***	0.039	0.482	0.448	0.049	***
Mixture	0.038	0.039	1.000	1.000	0.905	1.000	***	0.968	1.000	0.913	0.917	***
MSN	0.040	0.053	1.000	1.000	0.927	0.999	***	0.962	1.000	0.899	0.931	***
MVN	0.042	0.048	1.000	1.000	0.916	0.998	***	0.975	1.000	0.897	0.930	***
Tdist	0.045	0.231	0.999	1.000	0.870	0.962	***	0.296	0.721	0.412	0.279	***
							Model V′					
Chi	0.031	0.055	0.613	1.000	1.000	0.333	0.032	0.047	0.401	0.524	0.045	0.046
Mixture	0.042	0.049	0.983	1.000	1.000	0.916	1.000	0.816	0.990	0.996	0.715	0.914
MSN	0.050	0.047	0.928	1.000	1.000	0.932	1.000	0.719	0.994	0.997	0.732	0.935
MVN	0.046	0.047	0.967	1.000	1.000	0.935	1.000	0.778	0.995	0.997	0.734	0.932
Tdist	0.036	0.236	0.884	1.000	0.998	0.758	0.958	0.281	0.618	0.534	0.273	0.230

NOTE: The symbol *** here denotes that there is no value.

level s. We set the strength of each nonzero component of $\boldsymbol{\beta}_k$ to be equal to 1. For Models III' to V', we set the size of $\mathcal{M}_k$ to be $\lfloor \frac{s}{d} \rfloor$ where $k = 1, \cdots, d-1$, and $|\mathcal{M}_d| = s - \sum_{k=1}^{d-1} |\mathcal{M}_k|$. Due to page limit, we select only one representative variable for each index U_k and report its empirical rejection rate. We denote this rejection rate as "Index$k$". To reflect the Type-I error, we select one inactive variable randomly as "Inactive1".

To investigate the influence of the distributions of $\boldsymbol{X}$, we consider the following settings:

Table 4. Empirical rejection rate with respect to different sparsity level s.

	W_n	T^{NL}	W_n					T^{NL}				
s	Inactive1	Inactive1	Index1	Index2	Index3	Index4	Index5	Index1	Index2	Index3	Index4	Index5
							Model I′					
10	0.048	0.051	1.000	***	***	***	***	1.000	***	***	***	***
20	0.051	0.061	0.995	***	***	***	***	1.000	***	***	***	***
30	0.062	0.048	0.911	***	***	***	***	1.000	***	***	***	***
40	0.058	0.066	0.759	***	***	***	***	1.000	***	***	***	***
50	0.046	0.060	0.583	***	***	***	***	1.000	***	***	***	***
							Model II′					
10	0.060	0.048	1.000	1.000	***	***	***	1.000	0.056	***	***	***
20	0.043	0.043	0.975	0.965	***	***	***	0.962	0.042	***	***	***
30	0.064	0.051	0.796	0.695	***	***	***	0.803	0.044	***	***	***
40	0.045	0.056	0.612	0.429	***	***	***	0.583	0.063	***	***	***
50	0.040	0.051	0.460	0.261	***	***	***	0.467	0.058	***	***	***
							Model III′					
10	0.052	0.044	1.000	1.000	1.000	***	***	0.836	0.805	0.971	***	***
20	0.048	0.050	0.974	0.950	0.979	***	***	0.254	0.228	0.527	***	***
30	0.049	0.047	0.803	0.721	0.707	***	***	0.229	0.189	0.186	***	***
40	0.044	0.053	0.556	0.458	0.497	***	***	0.134	0.119	0.130	***	***
50	0.029	0.052	0.408	0.352	0.389	***	***	0.096	0.080	0.137	***	***
							Model IV′					
10	0.042	0.050	1.000	1.000	0.916	0.998	***	0.975	1.000	0.897	0.930	***
20	0.046	0.055	0.836	1.000	0.748	0.609	***	0.083	0.912	0.285	0.068	***
30	0.036	0.050	0.538	1.000	0.572	0.208	***	0.056	0.696	0.156	0.053	***
40	0.028	0.058	0.310	1.000	0.483	0.120	***	0.047	0.423	0.116	0.045	***
50	0.045	0.054	0.216	0.984	0.381	0.072	***	0.049	0.338	0.083	0.049	***
							Model V′					
10	0.046	0.068	0.967	1.000	1.000	0.935	1.000	0.778	0.995	0.997	0.734	0.932
20	0.041	0.059	0.573	0.999	0.995	0.377	0.181	0.091	0.822	0.844	0.086	0.058
30	0.047	0.046	0.354	0.999	0.948	0.185	0.033	0.059	0.586	0.520	0.055	0.034
40	0.043	0.060	0.237	0.986	0.863	0.102	0.045	0.053	0.382	0.279	0.042	0.054
50	0.046	0.047	0.174	0.979	0.764	0.077	0.047	0.046	0.271	0.206	0.049	0.050

NOTE: The symbol *** here denotes that there is no value.

(1) Multivariate normal distribution (MVN): $\boldsymbol{X} \sim \mathbf{MVN}(\mathbf{0}, \boldsymbol{\Sigma}_1)$, where $\boldsymbol{\Sigma}_1 = (\sigma_{1ij})$ with $\sigma_{1ij} = 0.5^{|i-j|}$, a covariance matrix with AR(1) structure.

(2) Multivariate T distribution (Tdist) with degrees of freedom 5 and covariance matrix $\boldsymbol{\Sigma}_1$.

(3) Chi-squared distribution (Chi): $\boldsymbol{X}^\top = (X_1, \dots, X_p)$, where $X_1, \dots, X_p$ are iid $\sim \chi_5^2 - 5$.

(4) Mixture normal distribution (Mixture): $\boldsymbol{X} \sim 0.2\mathbf{MVN}(\mathbf{0}, \boldsymbol{\Sigma}_1) + 0.4\mathbf{MVN}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2) + 0.4\mathbf{MVN}(\boldsymbol{\mu}_3, \boldsymbol{\Sigma}_3)$, where $\boldsymbol{\Sigma}_j$ satisfies AR(1) structure with correlation coefficients of 0.5, 0.2, and 0.8, respectively. $\boldsymbol{\mu}_2 = \Big(\underbrace{-1, \dots, -1}_{6}, 0, \dots, 0\Big)$ and $\boldsymbol{\mu}_3 = \Big(\underbrace{1, \dots, 1}_{6}, 0, \dots, 0\Big)$.

(5) Multivariate Skew-normal distribution (MSN, Azzalini and Capitanio 1999) with location parameter $\mathbf{0} \in \mathbb{R}^p$, scale parameter $\boldsymbol{\Sigma}_1$, and skewness parameter $(1, 1, 0, 0, 1, 1, 0, \dots, 0)$.

For distributions in (1)–(2), the linearity condition (LC) is satisfied, whereas it is violated for distribution specified in (3)-(5) (Guan, Xie, and Zhu 2017). Therefore, we can investigate whether our method can also perform well under non-elliptical distributions. We directly apply B-spline transformation, set $h = 7$, $(n, p) = (400, 2000)$ and $s = 10$ here.

The simulation results are summarized in Table 3. First, we note that our method controls the Type-I error well for all distributions. However, when $\boldsymbol{X}$ follows a multivariate T distribution, the Type-I error of T^{NL} is inflated to more than 0.2 under Model III'-V'. Regarding the power of our method, we observe that different distributions of $\boldsymbol{X}$ can result in different power levels for the same model. Specifically, the powers under the Chi-squared are relatively lower compared with the other distributions. For instance, in Model IV' and for Index4, the power is only 0.219 when $\boldsymbol{X}$ follows the Chi-squared distribution, whereas the power is greater than 0.962 for the other distributions. This also illustrates that our method can achieve high power even when the linearity condition is violated, as in the cases of MSN and Mixture distributions. Furthermore, our method consistently outperforms T^{NL} for all models in this simulation.

Table 5. Probe ID selected.

Probe ID	p-value	Probe ID	p-value
1382263_at	2.34×10^{-8}	1380883_at	1.62×10^{-4}
1369453_at	3.28×10^{-5}	1384488_at	1.66×10^{-4}
1377187_at	4.41×10^{-5}	1382517_at	1.77×10^{-4}
1390301_at	6.13×10^{-5}	1370857_at	2.08×10^{-4}
1379597_at	6.15×10^{-5}	1385543_at	2.43×10^{-4}
1376303_a_at	1.43×10^{-4}	1370551_a_at	2.91×10^{-4}

5.2.2. *Impact of Sparsity Level*

In this section, we investigate the effect of different sparsity levels. We consider the same data generation models as in Section 5.2.1, with $\boldsymbol{X} \sim \mathbf{MVN}(\mathbf{0}, \boldsymbol{\Sigma}_1)$, $(n, p) = (400, 2000)$, B-spline transformation and $h = 7$. Let the sparsity level s be $10, 20, 30, 40,$ and 50.

The simulation results for different sparsity levels are presented in Table 4. Both W_n and T^{NL} are able to control the Type-I error under the nominal level for all sparsity levels considered ($s = 10, 20, 30, 40, 50$). Comparing the performance of W_n and T^{NL} for the same sparsity level, we consistently observe that our proposed method is generally more powerful than T^{NL}, even for large sparsity levels such as $s = 50$.

5.3. A Real Data Example

We illustrate the proposed procedures by an empirical analysis of the data studied in Scheetz et al. (2006), Huang, Horowitz, and Wei (2010) and Li, Liu, and Lou (2017). This dataset contains 120 twelve-week old male rats from which over 18976 different probes (genes) from eye tissue were measured and can be viewed as "sufficient variables." The intensity values were normalized using the RMA (robust multi-chip averaging, Bolstad et al. 2003) method to obtain summary expression values for each probe. Our goal is to identify the probes that could significantly impact the gene TRIM32, which was previously reported in Chiang et al. (2006) as a potential cause of Bandet-Biedl syndrome, a genetically heterogeneous disease of multiple organ systems including the retina.

Following the procedure outlined in Huang, Ma, and Zhang (2008), we select 3000 probes with large variances from the 18976 probes. It is expected that only a few of these genes are related to TRIM32. We apply the newly proposed procedures to test whether the jth gene is associated with TRIM32, where we apply B-spline transformation with $h = 5$. In the implementation of the proposed FDR control procedure, we choose the tuning parameter λ for every probe using the 10-folds cross-validation. More details are given in Section S.4.5 in the supplementary material.

The proposed FDR control procedure identifies 12 probes at the FDR level 0.05. The IDs of identified probes and their p-values are listed in Table 5. Of the 12 probes selected by our proposed procedures, two probes (1382263_at and 1371187_at) were also identified by Huang, Horowitz, and Wei (2010), while one probe (1369453_at) was selected by Huang, Ma, and Zhang (2008). Additionally, two other probes (1380883_at and 1382517_at) were also selected in the work of Wang, Wu, and Li (2012). Due to the complexity of gene microarray data, it is not surprised that different approaches detect different important gene sets. Based on these repeated findings, we are more confident that the probes selected by the proposed procedures provide useful information for further biological study.

6. Conclusions and Discussions

In this article, we have developed model-free inference procedures for high-dimensional data by using a new reformation within sufficient dimension reduction framework and the orthogonalization technique. The proposed test statistic is shown to asymptotically follow χ^2 distribution under the null hypothesis, and a noncentral χ^2 distribution under local alternative hypotheses. The FDR control for a large-scale multiple testing problem using the proposed test to identify important predictors is also studied. We introduce the maximum correlation coefficient to quantify the dependence, and develop a truncated BH method to control the FDR. We show that the procedure is valid for controlling FDR under some mild conditions.

Supplementary Materials

The proofs of the main theorems, Proposition 2 and their related technical lemmas, and some additional simulation results are included in the online supplementary materials.

Acknowledgments

The authors are grateful to the Editor, the Associate Editor and referees for their constructive comments and suggestions that led to significant improvement of an early article.

Data Availability Statement

All numerical studies were conducted by using R code. The data and R code are available at webpage: *https://github.com/multizhang288/Model-Free-Inference*.

Disclosure Statement

The authors report there are no competing interests to declare.

Funding

X. Guo's research was supported by National Key R&D Program of China (grant nos. 2023YFA1008702, 2023YFA1011100) and National Natural Science Foundation of China (NNSF) grant (nos. 12322112, 12071038). R. Li's research was supported by National Institutes of Health grants R01AI136664 and R01AI170249. C. Zou's research was supported by the China National Key R&D Program (grant nos. 2022YFA1003703, 2022YFA1003800) and NNSF of China grants (nos. 11925106, 12231011, 11931001, 11971247, 12226007).

References

Azzalini, A., and Capitanio, A. (1999), "Statistical Applications of the Multivariate Skew Normal Distribution," *Journal of the Royal Statistical Society*, Series B, 61, 579–602. [195]

Barber, R. F., and Candès, E. J. (2015), "Controlling the False Discovery Rate via Knockoffs," *The Annals of Statistics*, 43, 2055–2085. [190]

Barber, R. F., Candes, E. J., and Samworth, R. J. (2020), "Robust Inference with Knockoffs," *Annals of Statistics*, 48, 1409–1431. [190]

Belloni, A., Chernozhukov, V., Chetverikov, D., and Wei, Y. (2018), "Uniformly Valid Post-regularization Confidence Regions for Many Functional Parameters in z-estimation Framework," *Annals of Statistics*, 46, 3643–3675. [189]

Belloni, A., Chernozhukov, V., and Kato, K. (2015), "Uniform Post-Selection Inference for Least Absolute Deviation Regression and other z-estimation Problems," *Biometrika*, 102, 77–94. [189]

Benjamini, Y., and Hochberg, Y. (1995), "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing," *Journal of the Royal statistical society*, Series B, 57, 289–300. [187]

Benjamini, Y., and Yekutieli, D. (2001), "The Control of the False Discovery Rate in Multiple Testing under Dependency," *Annals of Statistics*, 29, 1165–1188. [187]

Bolstad, B. M., Irizarry, R. A., Åstrand, M., and Speed, T. P. (2003), "A Comparison of Normalization Methods for High Density Oligonucleotide Array Data based on Variance and Bias," *Bioinformatics*, 19, 185–193. [196]

Cai, T., Cai, T. T., Liao, K., and Liu, W. (2019), "Large-Scale Simultaneous Testing of Cross-Covariance Matrices with Applications to Phewas," *Statistica Sinica*, 29, 983–1005. [187]

Candes, E., Fan, Y., Janson, L., and Lv, J. (2018), "Panning for Gold: 'model-x' Knockoffs for High Dimensional Controlled Variable Selection," *Journal of the Royal Statistical Society*, Series B, 80, 551–577. [186,187,190,193]

Chiang, A. P., Beck, J. S., Yen, H.-J., Tayeh, M. K., Scheetz, T. E., Swiderski, R. E., Nishimura, D. Y., Braun, T. A., Kim, K.-Y. A., Huang, J., et al. (2006), "Homozygosity Mapping with SNP Arrays Identifies trim32, an e3 ubiquitin ligase, as a Bardet–Biedl Syndrome Gene (bbs11)," *Proceedings of the National Academy of Sciences*, 103, 6287–6292. [196]

Cook, R. D. (2004), "Testing Predictor Contributions in Sufficient Dimension Reduction," *The Annals of Statistics*, 32, 1062–1092. [187]

Cook, R. D., and Forzani, L. (2008), "Principal Fitted Components for Dimension Reduction in Regression," *Statistical Science*, 23, 485–501. [188]

Cook, R. D., and Ni, L. (2006), "Using Intraslice Covariances for Improved Estimation of the Central Subspace in Regression," *Biometrika*, 93, 65–74. [188]

Ding, S., Qian, W., and Wang, L. (2020), "Double-Slicing Assisted Sufficient Dimension Reduction for High-Dimensional Censored Data," *The Annals of Statistics*, 48, 2132–2154. [187,188]

Dong, Y., Yang, C., and Yu, Z. (2016), "On Permutation Tests for Predictor Contribution in Sufficient Dimension Reduction," *Journal of Multivariate Analysis*, 149, 81–91. [187]

Fan, J., and Li, R. (2001), "Variable Selection via Nonconcave Penalized Likelihood and its Oracle Properties," *Journal of the American Statistical Association*, 96, 1348–1360. [191]

Fan, J., Li, R., Zhang, C.-H., and Zou, H. (2020), *Statistical Foundations of Data Science*, Boca Raton, FL: Chapman and Hall/CRC. [186,189,190]

Fang, E. X., Ning, Y., and Li, R. (2020), "Test of Significance for High-Dimensional Longitudinal Data," *Annals of Statistics*, 48, 2622–2645. [186]

Fang, E. X., Ning, Y., and Liu, H. (2017), "Testing and Confidence Intervals for High Dimensional Proportional Hazards Models," *Journal of the Royal Statistical Society*, Series B, 79, 1415–1437. [186]

Fung, W. K., He, X., Liu, L., and Shi, P. (2002), "Dimension Reduction based on Canonical Correlation," *Statistica Sinica*, 12, 1093–1113. [188]

Guan, Y., Xie, C., and Zhu, L. (2017), "Sufficient Dimension Reduction with Mixture Multivariate Skew-Elliptical Distributions," *Statistica Sinica*, 27, 335–355. [187,195]

Huang, J., Horowitz, J. L., and Wei, F. (2010), "Variable Selection in Nonparametric Additive Models," *Annals of Statistics*, 38, 2282–2313. [196]

Huang, J., Ma, S., and Zhang, C.-H. (2008), "Adaptive Lasso for Sparse High-Dimensional Regression Models," *Statistica Sinica*, 18, 1603–1618. [196]

Javanmard, A., and Montanari, A. (2014), "Confidence Intervals and Hypothesis Testing for High-Dimensional Regression," *The Journal of Machine Learning Research*, 15, 2869–2909. [186]

Li, B. (2018), *Sufficient Dimension Reduction: Methods and Applications with R*, Boca Raton, FL: CRC Press. [186,187,188]

Li, K.-C. (1991), "Sliced Inverse Regression for Dimension Reduction," *Journal of the American Statistical Association*, 86, 316–327. [188]

Li, R., Liu, J., and Lou, L. (2017), "Variable Selection via Partial Correlation," *Statistica Sinica*, 27, 983–996. [196]

Liu, W. (2013), "Gaussian Graphical Model Estimation with False Discovery Rate Control," *The Annals of Statistics*, 41, 2948–2978. [187,191]

Luo, W. (2022), "On Efficient Dimension Reduction with Respect to the Interaction between Two Response Variables," *Journal of the Royal Statistical Society*, Series B, 84, 269–294. [187]

Mai, Q., He, D., and Zou, H. (2022), "Coordinatewise Gaussianization: Theories and Applications," *Journal of the American Statistical Association*, 118, 2329–2343. [188]

Ning, Y., and Liu, H. (2017), "A General Theory of Hypothesis Tests and Confidence Regions for Sparse High Dimensional Models," *The Annals of Statistics*, 45, 158–195. [186,189,190,192]

Qian, W., Ding, S., and Cook, R. D. (2019), "Sparse Minimum Discrepancy Approach to Sufficient Dimension Reduction with Simultaneous Variable Selection in Ultrahigh Dimension," *Journal of the American Statistical Association*, 114, 1277–1290. [187,188]

Scheetz, T. E., Kim, K.-Y. A., Swiderski, R. E., Philp, A. R., Braun, T. A., Knudtson, K. L., Dorrance, A. M., DiBona, G. F., Huang, J., Casavant, T. L., et al. (2006), "Regulation of Gene Expression in the Mammalian Eye and its Relevance to Eye Disease," *Proceedings of the National Academy of Sciences*, 103, 14429–14434. [196]

Shao, Y., Cook, R. D., and Weisberg, S. (2007), "Marginal Tests with Sliced Average Variance Estimation," *Biometrika*, 94, 285–296. [187]

Shi, C., Song, R., Chen, Z., and Li, R. (2019), "Linear Hypothesis Testing for High Dimensional Generalized Linear Models," *The Annals of Statistics*, 47, 2671–2703. [186]

Storey, J. D. (2002), "A Direct Approach to False Discovery Rates," *Journal of the Royal Statistical Society*, Series B, 64, 479–498. [187]

Sun, Q., and Zhang, H. (2021), "Targeted Inference Involving High-Dimensional Data Using Nuisance Penalized Regression," *Journal of the American Statistical Association*, 116, 1472–1486. [186]

Tibshirani, R. (1996), "Regression Shrinkage and Selection via the Lasso," *Journal of the Royal Statistical Society*, Series B, 58, 267–288. [191]

Van de Geer, S., Bühlmann, P., Ritov, Y., and Dezeure, R. (2014), "On Asymptotically Optimal Confidence Regions and Tests for High-Dimensional Models," *The Annals of Statistics*, 42, 1166–1202. [186]

Wang, L., Wu, Y., and Li, R. (2012), "Quantile Regression for Analyzing Heterogeneity in Ultra-High Dimension," *Journal of the American Statistical Association*, 107, 214–222. [196]

Weng, J., and Yin, X. (2022), "A Minimum Discrepancy Approach with Fourier Transform in Sufficient Dimension Reduction," *Statistica Sinica*, 32, 2381–2403. [188]

Xia, Y., Cai, T. T., and Li, H. (2018), "Joint Testing and False Discovery Rate Control in High-Dimensional Multivariate Regression," *Biometrika*, 105, 249–269. [190,191]

Yin, X., and Cook, R. D. (2002), "Dimension Reduction for the Conditional kth Moment in Regression," *Journal of the Royal Statistical Society*, Series B, 64, 159–175. [188]

Yin, X., and Li, B. (2011), "Sufficient Dimension Reduction based on an Ensemble of Minimum Average Variance Estimators," *The Annals of Statistics*, 39, 3392–3416. [188]

Ying, C., and Yu, Z. (2022), "Fréchet Sufficient Dimension Reduction for Random Objects," *Biometrika*, 109, 975–992. [187]

Yu, X., Yao, J., and Xue, L. (2022), "Nonparametric Estimation and Conformal Inference of the Sufficient Forecasting with a Diverging Number of Factors," *Journal of Business & Economic Statistics*, 40, 342–354. [188]

Yu, Z., and Dong, Y. (2016), "Model-Free Coordinate Test and Variable Selection via Directional Regression," *Statistica Sinica*, 26, 1159–1174. [187,188]

Zeng, J., Mai, Q., and Zhang, X. (2024), "Subspace Estimation with Automatic Dimension and Variable Selection in Sufficient Dimension Reduction," *Journal of the American Statistical Association*, 1–13. [187,188]

Zhang, C.-H., and Zhang, S. S. (2014), "Confidence Intervals for Low Dimensional Parameters in High Dimensional Linear Models," *Journal of the Royal Statistical Society*, Series B, 76, 217–242. [186,192]